Robust Tip-Angle Tracking of a Flexible Link: A Comparison of LQG, Output-Feedback MPC, and H_∞ Control

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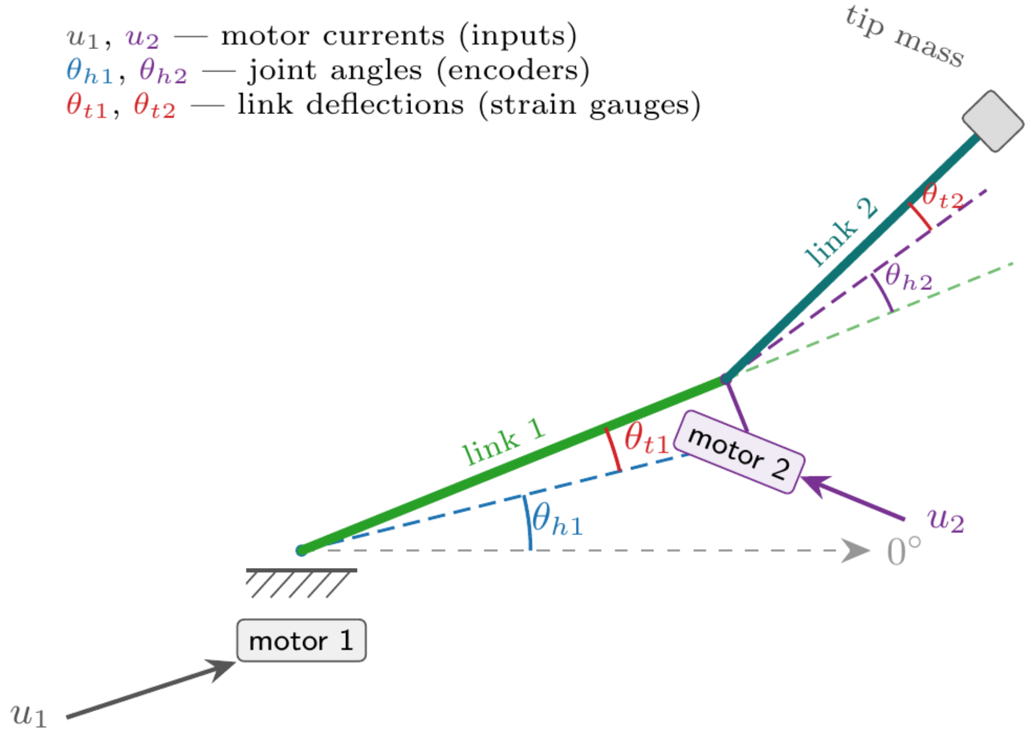

Fig. 1: Stage-1 variables. Input: motor current u. Measured: hub angle θ_h (encoder) and link deflection θ_t (strain gauge). The controlled output is the absolute tip angle $\theta_{\text{tip}} = \theta_h + \theta_t$.

$$A = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & K_s/J_{11} & -B_{11}/J_{11} & 0 \\ 0 & -\frac{J_{11}+J_{12}}{J_{11}J_{12}}K_s & B_{11}/J_{11} & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 0 \\ K_{tg}/J_{11} \\ -K_{tg}/J_{11} \end{pmatrix} \tag{2}$$

I. The Laboratory Model and the Control Task

A. The apparatus

The Quanser 2-DOF Serial Flexible Link is a planar two-link manipulator whose links are thin beams that visibly bend under motion. Each joint is driven by a DC motor through; each link carries a base strain gauge that measures its elastic deflection, and each joint carries an optical encoder. This project addresses the inner link only (the outer link and its motor are locked and treated as a rigid tip load), which already exhibits the essential difficulty of the full machine, a lightly-damped structural resonance in the loop, while remaining a single-input model suitable for a controlled comparison.

Fig. 1 defines the geometry. The motor applies a torque at the hub; the hub rotates by the encoder angle θ_h, and the beam bends by the additional strain-gauge angle θ_t, so the quantity a downstream task actually cares about (the *absolute tip angle*) is $\theta_{\text{tip}} = \theta_h + \theta_t$.

B. Mathematical model

Lumping the link as a two-inertia system (hub inertia J_{11}, tip inertia J_{12}) joined by a torsional spring of stiffness K_s and hub damping B_{11}, the linearised Stage-1 dynamics in the state $x = \left[\theta_h, \theta_t, \dot{\theta}_h, \dot{\theta}_t\right]^\top$ are $\dot{x} = Ax + Bu$ with A, B as in (2), where $K_{tg} = 0.75 K_m K_g$ is the motor-torque constant reflected through the gearbox. The controlled output and the two measurements are

$$y_{\text{ctrl}} = \underbrace{\begin{pmatrix} 1 & 1 & 0 & 0 \end{pmatrix}}_{C_{\text{tip}}} x, \quad y_{\text{meas}} = \underbrace{\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}}_{C_{\text{pos}}} x. \tag{1}$$

The nominal parameters (from the Quanser configuration files) and the resulting modal properties are collected in Table I. Two facts drive every design decision below. First, A has a *rigid-body integrator*: a pole at exactly the origin (rotary-arm torque integrates to angle), which is what lets a setpoint be tracked with zero steady-state error. Second, the structural mode at $\omega_n = 16.9$ rad/s (2.69 Hz) is *very lightly damped* ($\zeta = 0.33$): in open loop the tip rings for seconds, and any controller that pushes loop gain through this frequency risks exciting it.

TABLE I: NOMINAL STAGE-1 PARAMETERS AND MODAL PROPERTIES.

Symbol	Quantity	Value	Unit
K_s	link torsional stiffness	39.96	$\mathrm{N \cdot m/rad}$
J_{11}	hub equiv. inertia	0.0635	$\mathrm{kg \cdot m^2}$
J_{12}	tip equiv. inertia	0.1704	$\mathrm{kg \cdot m^2}$
B_{11}	hub viscous damping	4.0	$\mathrm{N \cdot m \cdot s/rad}$
K_{tg}	motor const. @ load	8.925	$\mathrm{N \cdot m/A}$
K_g	gear ratio	100	–
poles	eigenvalues of A	$0; -51.8; -5.58 \pm 15.9j$	
ω_n	flexible-mode freq.	16.9	rad/s
ζ	flexible-mode damping	0.33	–

C. Variables: inputs, states, outputs

The two positions θ_h, θ_t are the *only* signals available for feedback (as on the real rig), matching (1); the two velocities are estimated. Sampling is at $T_s = 10$ ms. Table II lists every variable with its operating range and unit.

TABLE II: INPUT, STATE, AND OUTPUT VARIABLES. CURRENT IS HARD-LIMITED AT 0.94 A IN FIRMWARE; SIMULATIONS CLIP AT $\pm u_{\max}$. THE 5° DEFLECTION LIMIT IS A SAFETY TRIP; CONTROLLERS TARGET A TIGHTER $\approx 2°$. VELOCITIES ARE RECONSTRUCTED BY THE OBSERVER.

Symbol	Meaning	Range	Unit
Input (manipulated)			
u	motor current command	$\lvert u \rvert \le 0.94$	A
States			
θ_h	hub joint angle	± 30 cmd	deg
θ_t	link tip deflection	$\lvert \theta_t \rvert \le 5$	deg
$\dot\theta_h$	hub rate	est.	deg/s
$\dot\theta_t$	deflection rate	est.	deg/s
Measured outputs			
$y_1 = \theta_h$	hub encoder	res. $8.8e^{-4}$	deg
$y_2 = \theta_t$	base strain gauge	± 5	deg
Controlled output			
θ_{tip}	$= \theta_h + \theta_t$	± 30	deg

D. Characterisation of uncertainty

The nominal model in (2) is not an exact model. This makes it a good fit for testing robustness of a controller designed on the theoretical model. The main differences are:

- Parametric stiffness K_s: The resonance measured on the rig (≈ 1.8 Hz, §6) sits *below* the modelled 2.69 Hz. We bracket this with a multiplier $K_s \in [0.3, 1.5] \times$ nominal.
- Unmodelled dynamics: The single-mode model omits higher bending modes and any actuator/ communication lag. These are unstructured, so no parameter in (2) represents them. I test two representatives in §4: a residual second flexible mode (spillover) and a pure input delay (phase lag).

E. Control requirements

The task is to drive the absolute tip angle θ_{tip} to a commanded set point (steps up to $\pm 30°$), subject to:

1) Zero steady-state error to a step.
2) $\lvert u \rvert \le u_{\max}$ A actuator limit at all times.
3) Keeping the tip deflection small, $\lvert \theta_t \rvert \lesssim 2°$, to bound beam stress.
4) Settle the lightly-damped mode quickly, no sustained ringing.
5) Remain stable under the uncertainties above, especially the unstructured high-frequency dynamics.

II. THE THREE CONTROL METHODS

All three controllers share the *same* nominal discrete model (A_d, B_d from a zero-order-hold of (2) at $T_s = 10$ ms), the *same* output set C_{pos}, and (to achieve a fair comparison) the *same* state estimator. Only the control law differs. Where a controller needs the unmeasured velocities, they come from one common steady-state Kalman predictor

$$\hat{x}_{k+1} = A_d \hat{x}_k + B_d u_k + L\big(y_k - C_{\text{pos}} \hat{x}_k\big), \tag{3}$$

with L from `dlqe` on process covariance $Q_n = q_n B_d B_d^\top$ ($q_n = 10$) and measurement covariance $R_n = r_n I$ ($r_n = 10^{-6}$, i.e. the positions are trusted).

A. LQG — linear–quadratic Gaussian

The simplest controller of the bunch, which will serve as a baseline benchmark. An infinite-horizon LQR gain K is computed once by solving the discrete algebraic Riccati equation (`dlqr`) for the cost

$$J = \sum_{k=0}^{\infty} (x_k - x_{\text{ss}})^\top Q (x_k - x_{\text{ss}}) + R u_k^2, \tag{4}$$

with $Q = q_y C_{\text{tip}}^\top C_{\text{tip}} + 10^{-3} I$ ($q_y = 50$, i.e. penalise tip *angle* error) and $R = 0.5$. The control is static feedback on the estimate, with the reference injected as a state setpoint $x_{\text{ss}} = [r, 0, 0, 0]^\top$ (the plant's own integrator gives offset-free tracking):

$$u_k = \operatorname{sat}_{\pm u_{\max}}\big(-K(\hat{x}_k - x_{\text{ss}})\big). \tag{5}$$

LQG is fast and cheap, and on a well-modelled plant near-optimal — but its loop has *no guaranteed gain/phase margin*, a fact §4 shows clearly.

B. Output-feedback MPC

Same estimator, same Q, R, but the control is recomputed online by solving, at every step, a finite-horizon ($N_p = 40$, i.e. 0.4 s) constrained quadratic program seeded from the current estimate, (6). The terminal cost P is the LQR Riccati solution (so that the MPC is identical to LQG and to achieve a better comparison), and the deflection limit $\lvert \theta_t \rvert \le \theta_{\max}$ is imposed as a *soft* constraint (slack σ, penalty $\rho = 10^4$) to keep the QP always feasible. Solved with Clarabel. The distinguishing capability is the *state constraint*: MPC is the only one of the three that can actively cap the tip deflection.

C. H_∞ output feedback (Ch. 11–12)

A two-measurement mixed-sensitivity design synthesised with `hinfsyn` (Fig. 2). Two issues make this plant awkward and shape the design:

- The rigid-body pole on the imaginary axis breaks the synthesis rank conditions, so it is regularised off-

$$
\begin{aligned}
\min_{U,X,\sigma} \quad & \sum_{j=0}^{N_p-1} \Big[\big(x_j - x_{\text{ss}}\big)^{\top} Q \big(x_j - x_{\text{ss}}\big) + R u_j^2 + \rho \sigma_j \Big] + \big(x_{N_p} - x_{\text{ss}}\big)^{\top} P \big(x_{N_p} - x_{\text{ss}}\big) \\
\text{s.t.} \quad & x_{j+1} = A_d x_j + B_d u_j, \quad x_0 = \hat{x}_k, \\
& |u_j| \le u_{\max}, \quad |\theta_{t,j}| \le \theta_{\max} + \sigma_j, \quad \sigma_j \ge 0
\end{aligned}
\tag{6}
$$

axis, $A \leftarrow A - \varepsilon I$ ($\varepsilon = 10^{-3}$). But that same nudge hides the integrator from the synthesis, which then meets the tracking spec with feed-forward and zero DC feedback gain, this causes tracking with a large offset. We restore integral action explicitly with a tracking-error integrator state $\dot{x}_i = r - \theta_{\text{tip}}$ fed to the controller, recovering offset-free tracking.
- The regulated outputs are $z = [W_i x_i, \quad W_2 u]$. The control-effort weight W_2 is a high-pass with its corner placed *low*, at $\omega_c = 8$ rad/s — *below* the flexible mode — so the synthesis penalises high-frequency control effort and *rolls the loop off before the resonance*. This buys the robustness margin of §4 at the cost of a little nominal speed.

The synthesis returns a 6th-order controller with $\gamma = 0.31$. Because the integral augmentation adds a second imaginary-axis pole the synthesis is numerically delicate (about a quarter of the weight grid collapses to a degenerate controller), so the weights were difficult to select. Through trial-and-error I reached($\varepsilon = 10^{-3}$, $W_i = 1.0$, W_2 DC gain 0.1) but manual tuning proved unwieldy.

III. Nominal Performance on the Theoretical Model

On the nominal plant the three methods are, as expected, close — this is LQG/MPC's home turf, and H_∞'s robustness margin is invisible here. Fig. 3 overlays the closed-loop step to +20° and Table III quantifies it.

TABLE III: Nominal metrics (step to +20°). All three offset-free (steady-state error $< 0.01°$). LQG is fastest but its deflection (3.56°) exceeds the 2° limit; MPC trades 0.05 s of rise to hold it at exactly 2.00°; H_∞ is slowest but uses least current.

Method	**rise [s]**	**OS [%]**	**pk $|u|$ [A]**	**max $|\theta_t|$ [°]**
LQG	0.30	1.5	0.60	3.56
output-MPC	0.35	2.9	0.60	**2.00**
H_∞	0.49	0.7	0.46	2.25

The reading: *MPC's value is the state constraint*, not raw speed — pure input saturation does not separate it from a clipped LQR for this single move, but the deflection cap does, and only MPC can impose it. H_∞ looks merely "slower" here; the next section shows what that slowness is buying.

To see the loop-shaping explicitly, Fig. 4 plots the closed-loop sensitivity S and complementary sensitivity T of the H_∞ design. S rolls off to a small value at DC (offset-free tracking) and T rolls off *before* the flexible mode, with modest peaks — low peaks mean large stability margins, which is the currency spent in §4.

IV. Where H_∞ Earns Its Keep: Two Unstructured Uncertainties

The honest conclusion from §3 is that on a well-modelled, minimum-phase plant H_∞ has no visible advantage. Its margin only shows once the truth plant differs from the design model in ways *no quadratic cost accounts for*. We keep every controller exactly as designed (each still

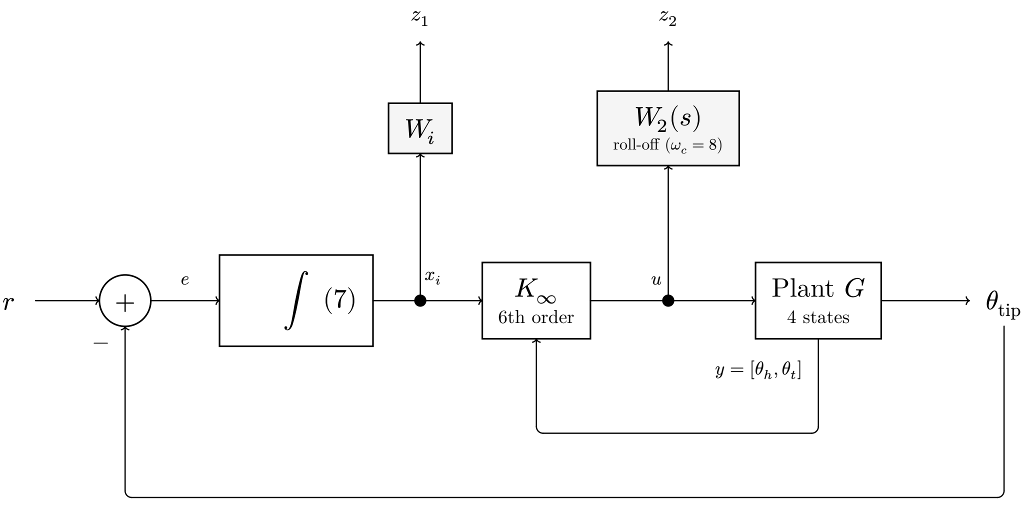

Fig. 2: Integral-augmented H_∞ output-feedback loop. The plant integrator is regularised off-axis for synthesis, so integral action is restored *explicitly*: the tracking error $e = r - \theta_{\text{tip}}$ is integrated to a state x_i that is handed to the controller alongside the two measurements $y = [\theta_h, \theta_t]$. The synthesis minimises the weighted regulated outputs $z_1 = W_i x_i$ (tracking) and $z_2 = W_2(s)u$ (control effort, rolled off above $\omega_c = 8$ rad/s).

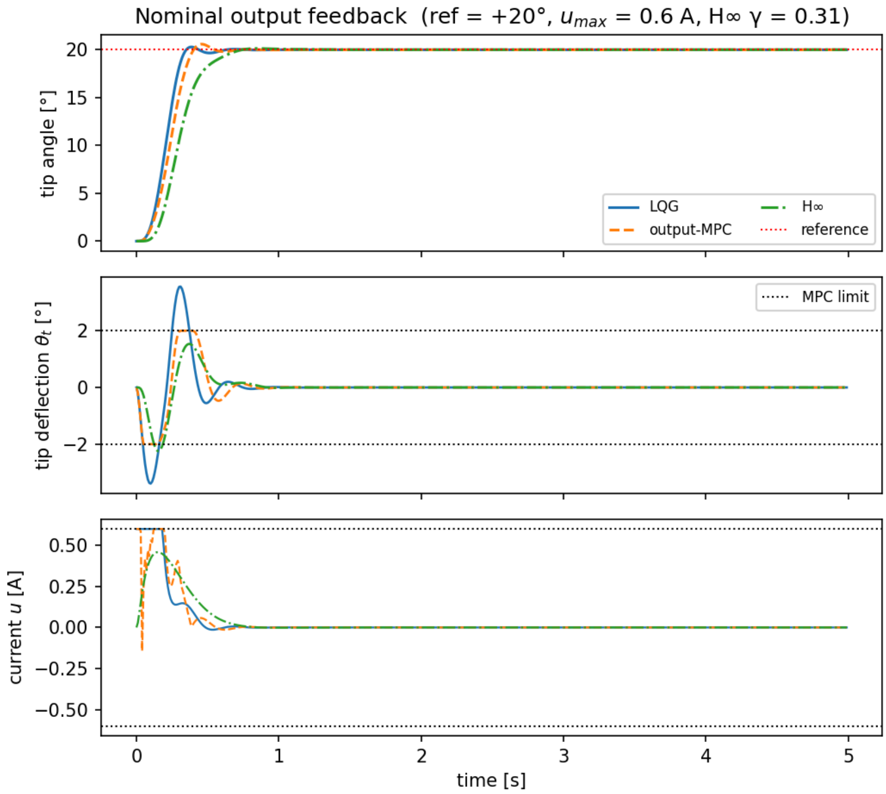

Fig. 3: Nominal closed-loop response to a +20° tip-angle step ($u_{\max} = 0.6$ A, MPC deflection limit 2°). Top: tip angle; middle: tip deflection θ_t with the MPC limit; bottom: motor current. All three track offset-free; only MPC holds the deflection at its constraint.

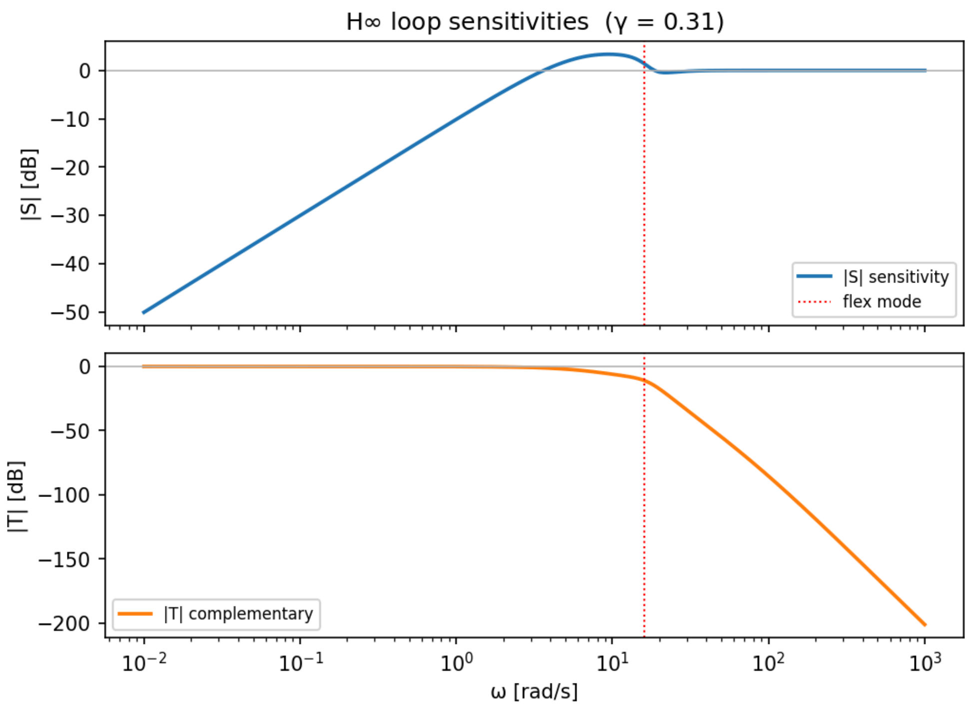

Fig. 4: Sensitivity S and complementary sensitivity T of the H_∞ loop (controlled output θ_{tip}). $|S(0)|$ small → offset-free; $|T|$ rolls off below the flexible mode (dotted) → no spillover; low $\|S\|_\infty, \|T\|_\infty$ peaks → margins.

uses the nominal four-state model and the same Kalman estimate) and perturb only the *simulated truth* plant.

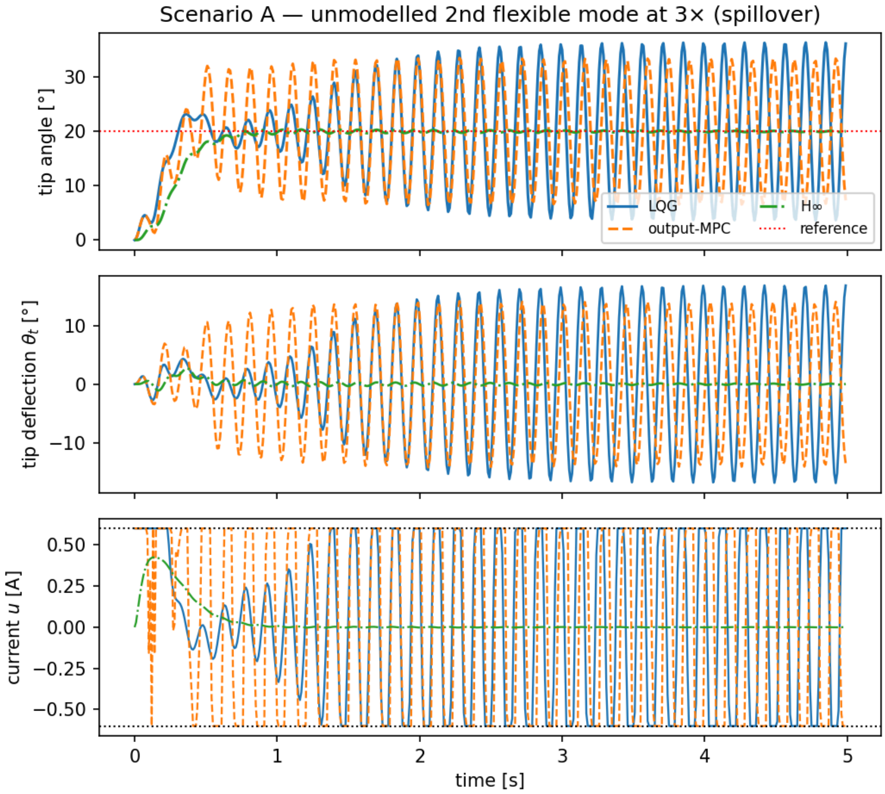

Fig. 5: Scenario A: with an unmodelled second flexible mode in the truth plant, nominal LQG/MPC excite it and ring (control spillover) while the rolled-off H_∞ loop tracks cleanly.

A. Scenario A — unmodelled second flexible mode (spillover)

A real beam has more bending modes than the one-mode design model. We append a residual lightly-damped mode at $3\times$ the modelled frequency, coupled into the tip output. The nominal-designed *LQG and MPC push loop gain into the unmodelled mode and ring / destabilise* (control spillover — the closed-loop spectral radius exceeds 1), while the H_∞ loop, deliberately rolled off above the modelled mode, *stays clean*. Only an explicit high-frequency roll-off *guarantees* this; a quadratic cost provides none. See Fig. 5.

B. Scenario B — input delay (phase lag)

A few tens of milliseconds of actuator/communication delay — modelled as a discrete shift register at the plant input — eats phase margin. The margin-blind *LQG loses stability around* ≈ 50 *ms* (the textbook "LQG has no guaranteed margins" point), whereas H_∞, which bounds the sensitivity peaks, *tolerates substantially more* (80 ms+). See Fig. 6.

The price H_∞ pays for this is exactly the slower nominal response and the higher-order (6th vs static) controller seen in §3. *This is the central trade of the project*: LQG/MPC win on a model you trust; H_∞ wins when you do not.

V. Empirical Robustness Study: Parametric Perturbations

The unstructured uncertainties of §4 are where H_∞ is meant to shine. For completeness we also sweep the one *structured* uncertainty the model exposes — the link stiffness K_s, whose true value is the least certain parameter (§1.4). Holding each controller fixed at its nominal design, we re-simulate the $+20°$ step while scaling the truth-plant stiffness over $K_s \in [0.3, 1.5] \times$ nominal and record tracking error, overshoot, and peak deflection. See Fig. 7.

The instructive (and slightly counter-intuitive) result is that *this axis does not favour* H_∞: all three controllers remain robustly stable across the whole stiffness range, because K_s only moves the resonance, it does not introduce dynamics outside the model's structure. The integral action keeps LQG and MPC offset-free throughout; H_∞ too, apart from a small residual at the softest plants ($K_s \lesssim 0.4\times$ nominal), where its deliberate roll-off makes the loop slow enough that the integrator has not fully settled within the 5 s window. The differences are otherwise second-order. This is the complement of §4's lesson — robustness is not a scalar. A controller can be robust to large *parametric* error and yet fragile to small *unstructured* error (LQG, §4), or vice versa. Naming the uncertainty you actually face is the design's first job.

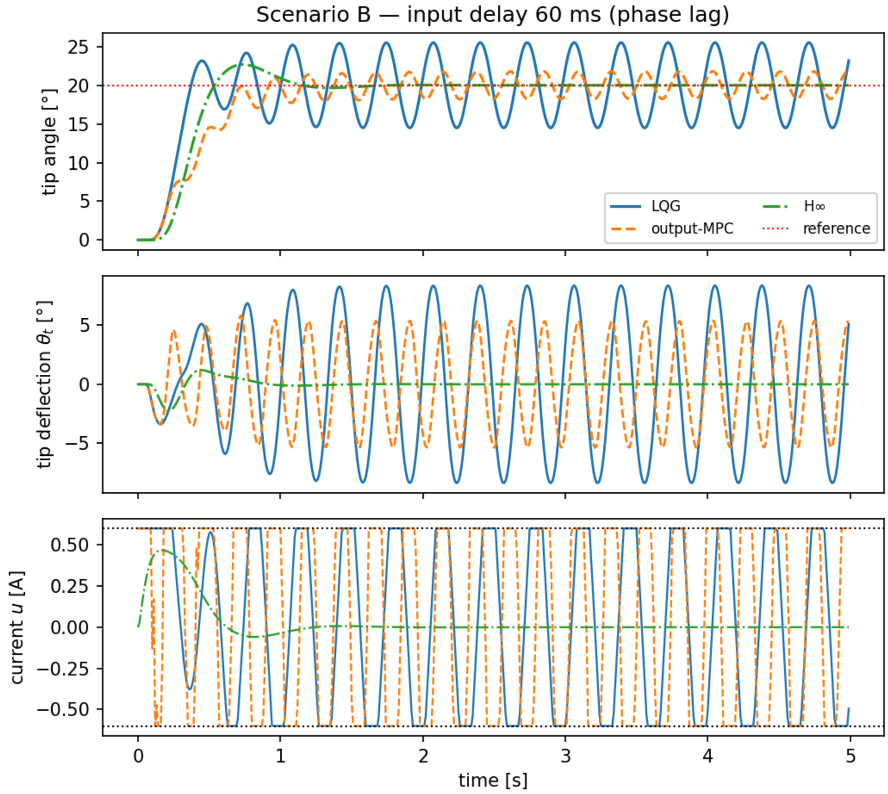

Fig. 6: Scenario B: a pure input delay. LQG destabilises near 50 ms; the H_∞ loop, with bounded $\|S\|_\infty$, keeps stable margins well past 80 ms.

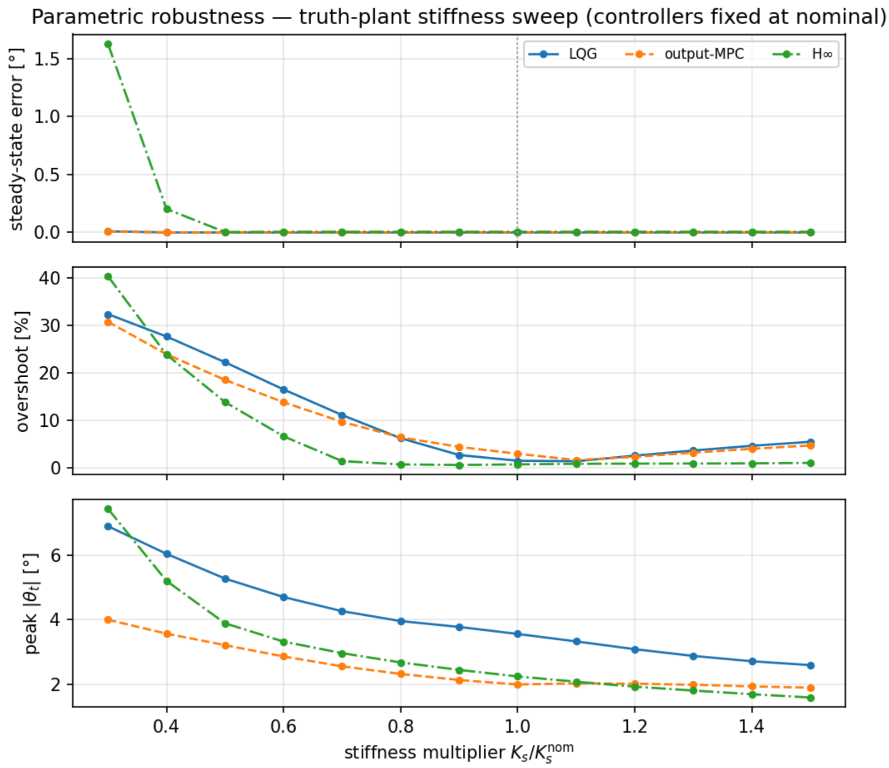

Fig. 7: Parametric robustness: steady-state error, overshoot, and peak deflection versus a truth-plant stiffness multiplier $K_s/K_s^{\mathrm{nom}} \in [0.3, 1.5]$, all controllers fixed at nominal design.

VI. Hardware Verification on the Physical Rig

Sections 3–5 are simulation. To verify that the conclusions survive contact with the apparatus, the two linear designs (LQG and H_∞) are deployed on the Quanser rig and their measured step responses compared against the Python prediction they were designed from. (MPC, which needs an online QP solver on the real-time target, was not deployed; it remains the simulation reference of §3.) The result is instructive: it does *not* simply confirm §4 — on the real plant the ranking *reverses*, for a reason that sharpens the project's thesis.

A. Deployment: the identical controller, on the rig

The rig's control loop is a Simulink/QUARC model (`q_2dsfl_pos_cntrl.slx`) running on a Q8-USB board at 500 Hz; the stock model uses Quanser's own continuous full-state-feedback LQR. To verify *our* designs we replace that single gain block and change nothing else in the safety/IO path. Each linear controller — the Kalman observer, the LQG gain (or the integral-augmented H_∞ dynamics) — folds into a *single* discrete state-space block, exported by `report/export_controllers.py` and verified there to reproduce the §3 simulation (H∞ to machine precision, LQG to $10^{-3\circ}$); `build_orr_models.m` drops it into the model automatically. Two points keep the deployment *faithful* rather than merely similar:

- *Same estimator in the loop.* The block takes only the two measured positions and reconstructs the velocities with the *same* Kalman predictor ((3)) used in simulation — the rig's own filtered derivatives are *not* used — so the loop transfer function matches the design, not an accidental variant of it.
- *Same numbers.* The gains are exported from the very script that produced Fig. 3, so there is no second re-tuning step on the rig.

B. Experimental protocol

Each controller drives the automated stair sequence of `run_lab_experiments.m` — steps to $\pm5, \pm10, \pm20, \pm30°$ at $u_{\max} = 0.6$ A — logging the full state at 500 Hz; each segment steps from the previous stair level, not from zero. The rig shapes every commanded step through a smoothing sigmoid, so `report/make_hw_figures.py` drives the Python prediction with the *logged* reference: the residual between measured and predicted then reflects *plant–model mismatch*, not the command shape. Fig. 8 and Fig. 9 show the +20° step; Table IV collects the measured-vs-predicted metrics.

TABLE IV: Hardware metrics, measured (m) vs. predicted (p), step to +20° (from `make_hw_figures.py`). LQG matches its prediction closely; H_∞ overshoots 5 × more than predicted. Across the ±20/30° steps the residual tip ring is 0.2–0.5° for LQG but 1.8–3.0° for H_∞.

Method	rise m/p [s]	OS m/p [%]	max $\lvert\theta_t\rvert$ [°]	sse [°]
LQG	0.76/0.70	3.5/0.0	**1.64**	0.04
H_∞	0.93/0.96	16.8/0.5	1.91	0.52

C. What the mismatch reveals: the ranking reverses

The rig is *not* the nominal model, so the value of this section is in the gap — and the gap is unequal between the two controllers.

LQG verifies cleanly. Measured rise tracks the prediction within ≈ 0.1 s across every amplitude, steady-state error stays under ≈ 0.6°, and the structural mode is actively damped to a 0.2–0.5° residual. The design transfers to hardware essentially as Fig. 3 promised.

H_∞, the simulation's robustness hero, is the one that struggles. It rings hard on the large steps (16–48% overshoot, 1.8–3° of sustained deflection at ≈ 1.8 Hz) and *stalls* on the small ones ($\pm5, +10°$ end 3–4.6° short — static friction the low-authority loop never overcomes). Two real-plant effects, both named in §1.4, drive this: hub *stiction* (a nonlinearity no linear design models) and a true resonance at ≈ 1.8 Hz — *not* the modelled 2.69 Hz, and right where the H_∞ loop was deliberately given little gain. Its roll-off, which bought margin against the *unstructured* high-frequency uncertainty of §4, leaves it without the authority to actively damp the (kicked) structural mode or to break stiction. LQG's higher loop gain — exactly the property §4 flagged as fragile to unstructured error — is what wins here.

This does not contradict §4; it *completes* it. H_∞ is robust to the uncertainty we simulated; LQG wins against the uncertainty the rig actually has. One honest caveat: the H_∞ weights were shaped around the *modelled* 2.69 Hz resonance, so part of its poor showing is a design tuned on a mis-located mode — which is itself the §4 lesson restated.

VII. Summary and Outlook

From one nominal model and one shared estimator we built three tip-angle trackers and compared them on identical terms:

- *All three* track the absolute tip angle offset-free, exploiting (LQG/MPC) or restoring (H_∞) the plant's rigid-body integrator.
- *MPC* is the only one that can enforce the tip-deflection limit; it holds $\lvert\theta_t\rvert$ at exactly 2° where LQG overshoots to 3.6°, at a cost of 0.05 s of rise.
- H_∞ is unremarkable on the nominal plant but is the only design that survives *unstructured* uncertainty — an unmodelled mode (spillover) and an input delay — because its loop is shaped to roll off below the resonance.

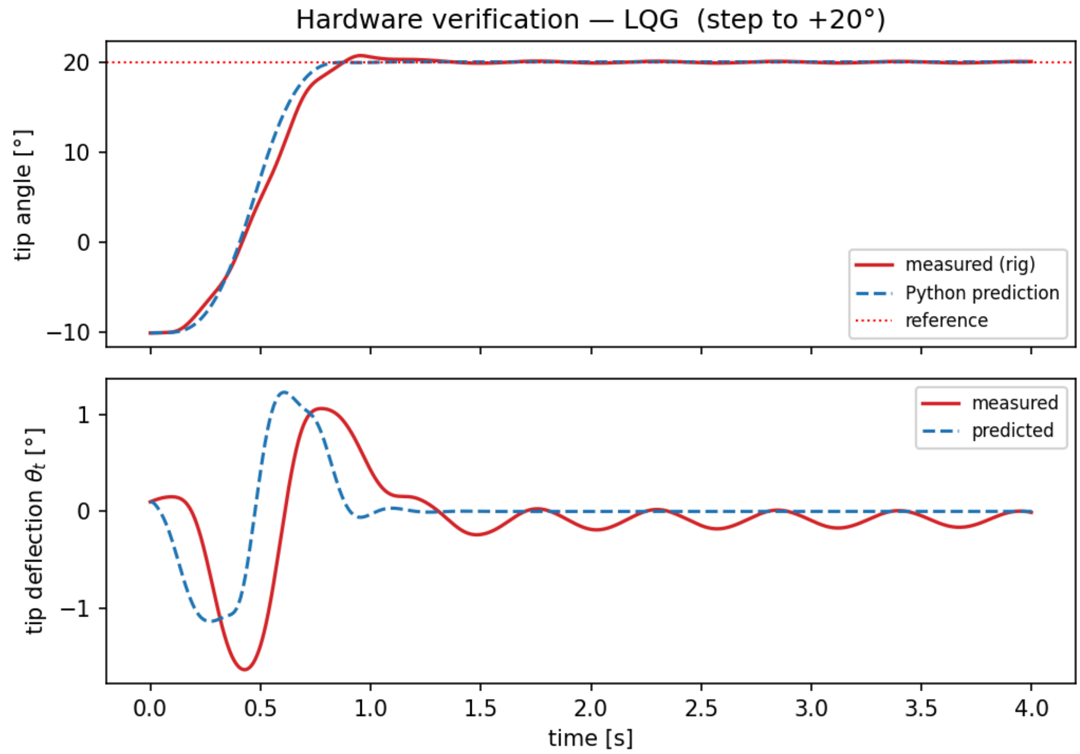

Fig. 8: LQG on the rig (red) vs. the logged-reference Python prediction (blue), +20° step. Angle and deflection track the design closely; the small residual ring (≈ 0.2°) is the lightly-damped structural mode, actively damped by the high-gain state feedback.

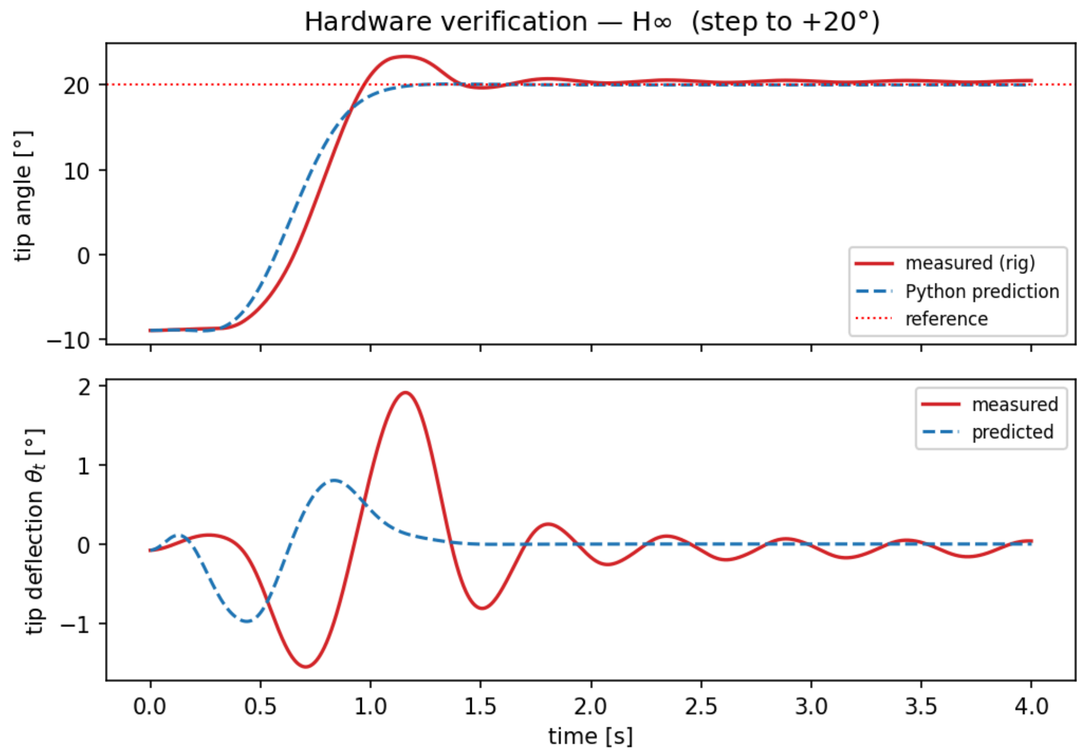

Fig. 9: H_∞ on the rig (red) vs. prediction (blue), +20°. The measured loop overshoots and rings at ≈ 1.8 Hz with 1.8–3° of deflection — far more than predicted — because its deliberately low-authority loop adds little active damping at the true resonance.

- The *parametric* stiffness sweep favours none of them: structured error is the easy case here.
- *On the real rig* (§6) the simulation ranking *reverses*: LQG tracks its prediction closely while H_∞ rings

and stalls, because the apparatus's dominant uncertainties — hub stiction and a resonance at $\approx$ 1.8 Hz rather than the modelled 2.69 Hz — are *not* the unstructured high-frequency dynamics H_∞ was shaped against, and its low-authority loop lacks the gain to damp them.

The message for design practice: there is no single "most robust" controller; the right choice follows from *which uncertainty dominates* — state constraints $\to$ MPC, unstructured high-frequency dynamics $\to$ H_∞, and (as the hardware shows) friction plus a mis-located resonance $\to$ the higher-authority LQG. The simulation and the rig do not disagree; they name *different* dominant uncertainties, and the right controller follows each.

Outlook. MPC was not deployed (its online QP needs a real-time solver on the target); closing that gap — explicit/move-blocking MPC on the rig, to test whether its deflection constraint also holds under stiction — is the natural next step, alongside re-shaping the H_∞ weights around the *measured* 1.8 Hz mode.

VIII. Appendix: Reproducibility and Figure Export

All designs and simulations live in `analysis/compare_methods_output.py` (an interactive marimo notebook). For the report, the script `report/make_figures.py` is a standalone, non-interactive port that designs the three controllers once and regenerates every figure of Table V into `report/figs/`. Edit its `CONFIG` block (reference, current limit, deflection limit, perturbation amounts), rerun, and recompile:

```
.venv/bin/python report/make_figures.py
cd report && typst compile --root ..
report.typ
```

TABLE V: Figures 1–5 generated by `make_figures.py` the hardware figures by `make_hw_figures.py`, into `report/figs/`.

Fig.	What it shows	Filename
1	nominal step ($u_{\max} = 0.6$, ref +20°, deflection limit 2°)	`compare_nominal.png`
2	H∞ sensitivities S, T (nominal weights)	`hinf_sensitivity.png`
3	Scenario A: unmodelled 2nd mode, $f_2 = 3\,\times$, coupling 1.0	`robust_spillover.png`
4	Scenario B: input delay $\approx$ 60 ms	`robust_delay.png`
5	K_s multiplier sweep 0.3 $\to$ 1.5	`robust_ks_sweep.png`
6,7	hardware: measured vs. predicted, LQG / H_∞	`hw_{lqg,hinf}.png`

Hardware pipeline (§6). `export_controllers.py` writes the controllers (and their verified single-block state-space realizations) to `orr_controllers.mat` `build_orr_models.m` generates the rig models `q_2dsfl_orr_{lqg,hinf}.slx` from the stock Quanser model; runs are logged by `run_lab_experiments.m` and overlaid against the logged-reference prediction by `make_hw_figures.py`. See `software/quanser_updated/WIRING.md`.

Environment: Python 3.12, `numpy`, `scipy`, `python-control` 0.10, `cvxpy` + Clarabel, `slycot`. Nominal model and parameters from the Quanser configuration files (`software/quanser_updated/config_2dslf.m`). Sampling $T_s = 10$ ms, horizon $N_p = 40$, $N_{\text{sim}} = 500$ steps. Rig: Q8-USB, 500 Hz, single-tasking.